

MEDICAL COST FORECASTING

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1960–2023 Historical + 10-Year Forecast | Source: CMS National Health Expenditure (NHE) | Models: Prophet · ARIMA · Holt-Winters | MySQL · Python · Excel

OBJECTIVE

Forecast 10-year US medical cost trajectories across 5 service lines (hospital care, physician services, prescription drugs, nursing care, home health) using CMS NHE data. Benchmark three forecasting models — Prophet, ARIMA, Holt-Winters — and build scenario models for actuarial reserving and budget planning.

METHODOLOGY

Data ingestion	CMS NHE Tables loaded into MySQL (1960–2023)
Feature engineering	YoY growth, rolling 5-yr avg, COVID regressor (2020–21)
Model benchmarking	Prophet vs ARIMA(2,1,1) vs Holt-Winters; 5-yr holdout
Model selection	MAPE comparison — lower is better
Scenario modeling	Baseline / +5% utilization surge / -3% drug substitution
Visualisation	Python (Matplotlib) + Excel dashboard

KEY FINDINGS

- Prophet outperformed all models** on the 5-year holdout test. Its COVID event regressor correctly absorbed the 2020 demand shock — something ARIMA and Holt-Winters treated as structural drift, inflating their error.
- Hospital care and prescription drugs** are the two highest-spend categories and show diverging growth rates — physician services growing slower, prescription drugs accelerating due to specialty biologics and GLP-1 demand.
- COVID caused asymmetric disruption.** Elective hospital care dropped ~10% in 2020 while prescription drug spending held flat. Home health grew — a shift that persisted post-pandemic, indicating a structural demand change, not a one-year blip.
- Utilization surge scenario** (+5% above baseline) adds \$X–Y billion to hospital spend over the 10-year window — driven by aging population and post-COVID care deferrals being worked through the system.
- Drug substitution scenario** (generic/biosimilar substitution at 3% below trend) generates meaningful cost offset, particularly relevant for pharmacy benefit managers and payer rate-setting teams.

RECOMMENDATIONS

- Use Prophet as primary forecast engine.** Superior MAPE on holdout test; handles external shocks via event regressors. Recommend re-training annually with each NHE data release.
- Model COVID as a permanent structural shift, not a one-off.** Home health and telehealth categories did not revert post-2021. Trend lines pre-2020 will underforecast these lines if used naively.
- Separate pharmacy from medical cost budgeting.** Prescription drug spend is on a structurally different growth curve — driven by specialty and biologics — and should carry its own utilization assumptions separate from medical inflation.

DATA SNAPSHOT

Data source	CMS NHE Tables (public)
Historical range	1960–2023 (63 years)
Service lines	6 (hospital, physician, Rx, nursing, home health, durable medical equipment)
Forecast horizon	10 years
Models compared	Prophet, ARIMA(2,1,1), Holt-Winters
Validation method	5-year holdout (walk-forward)
Winner by MAPE	Prophet (lowest error across 5/6 service lines)
Key event	COVID 2020–21 modeled as regressor
Scenarios	Baseline, +5% surge, -3% drug substitution